

ORF 524 – Midterm Exam

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Notes:

- Duration of examination: *4 hours*.
An additional 20 minutes are given to upload your solutions to Canvas.
- All parts are equally weighted (5 points).
Please provide as much detail as possible in your answers.
Answers without proper justification will not receive full credit.
- Please start each question on a separate page.
A single PDF file must be submitted to Canvas.
- Only two pages of personal notes are allowed during the examination.
Basic auxiliary information about standard distributions is provided at the end of the exam.
This is otherwise a closed-book and individual examination.
Do not consult textbooks, online materials, or other sources.
Do not consult with classmates, other people or in online forums.

1 Question 1: Probability Theory (25 points)

Let $(X_1, X_2, \dots, X_n)$ be independent (but not necessarily identically distributed) random variables with $\mu = \mathbb{E}[X_i]$ and $\sigma_i^2 = \mathbb{V}[X_i]$ for $i = 1, 2, \dots, n$ (all expectations are assumed to exist). A random variable Y is said to be *sub-Gaussian* if there is a positive parameter γ such that

$$\mathbb{E}[\exp(t(Y - \mathbb{E}[Y]))] \leq \exp\left(\frac{\gamma^2 t^2}{2}\right), \quad \text{for all } t \in \mathbb{R}.$$

We explore this property and use it to construct valid finite-sample and distribution-free confidence intervals for the common mean μ based on the sample average $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$.

1. Let $X_i \sim \text{Normal}(\mu, \sigma_i^2)$ for $i = 1, 2, \dots, n$. Show that $\bar{X}$ is sub-Gaussian, and identify its parameter γ .

- **Solution:** By properties of the Gaussian distribution, $\bar{X} = \text{Normal}(\mu, \bar{\sigma}^2/n)$, where $\bar{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n \sigma_i^2$. Using the moment generating function of the Gaussian distribution, the result follows with $\gamma = \bar{\sigma}/\sqrt{n}$.

2. Let X_i be sub-Gaussian with parameter γ_i for $i = 1, 2, \dots, n$.

- (a) Show that $\bar{X}$ is sub-Gaussian, and identify its parameter.

- **Solution:** By independence and sub-Gaussianity of X_i ,

$$\mathbb{E}[\exp(t(\bar{X} - \mu))] = \prod_{i=1}^n \mathbb{E}[\exp(t(X_i - \mu)/n)] \leq \prod_{i=1}^n \exp\left(\frac{\gamma_i^2 t^2}{2n^2}\right) = \exp\left(\frac{\tilde{\gamma}^2 t^2}{2n}\right),$$

with $\tilde{\gamma} = \sqrt{\frac{1}{n} \sum_{i=1}^n \gamma_i^2}$, and therefore $\bar{X}$ is sub-Gaussian with parameter $\gamma = \tilde{\gamma}/\sqrt{n}$.

- (b) Show that

$$\mathbb{P}[\bar{X} - \mu \geq x] \leq \inf_{t>0} \exp\left(\frac{\tilde{\gamma}^2 t^2}{2n} - tx\right) = \exp\left(-\frac{nx^2}{2\tilde{\gamma}^2}\right), \quad \text{for all } x \in \mathbb{R},$$

where $\tilde{\gamma} = \sqrt{\frac{1}{n} \sum_{i=1}^n \gamma_i^2}$. Use this result to prove the (two-sided) Hoeffding Inequality:

$$\mathbb{P}[|\bar{X} - \mu| \geq x] \leq 2 \exp\left(-\frac{nx^2}{2\tilde{\gamma}^2}\right).$$

- **Solution:** For any $t > 0$,

$$\begin{aligned} \mathbb{P}[\bar{X} - \mu > x] &= \mathbb{P}[\exp(t(\bar{X} - \mu)) \geq \exp(tx)] \leq \frac{\mathbb{E}[\exp(t(\bar{X} - \mu))]}{\exp(tx)} \\ &\leq \exp\left(\frac{\tilde{\gamma}^2 t^2}{2n} - tx\right). \end{aligned}$$

Therefore, Chernoff Inequality follows by applying $\inf_{t>0}$,

$$\mathbb{P}[\bar{X} - \mu > x] \leq \inf_{t>0} \exp\left(\frac{\tilde{\gamma}^2 t^2}{2n} - tx\right),$$

The value of t that minimizes the above expression is $t = \frac{xn}{\tilde{\gamma}}$, obtained by checking the FOC. This gives the equality

$$\inf_{t>0} \exp\left(\frac{\tilde{\gamma}^2 t^2}{2n} - tx\right) = \exp\left(-\frac{nx^2}{2\tilde{\gamma}^2}\right).$$

The two-sided Hoeffding Inequality now follows by the union probability bound.

3. Show that if a random variable W satisfies

$$\mathbb{P}[|W| \geq w] \leq a \exp\left(-\frac{bw^2}{c + dw}\right), \quad \text{for all } w > 0,$$

where $a, b > 0$ and $c, d \geq 0$, then

$$\mathbb{P}\left[|W| \leq \sqrt{\frac{c}{b} \ln\left(\frac{a}{\alpha}\right)} + \frac{d}{b} \ln\left(\frac{a}{\alpha}\right)\right] \geq 1 - \alpha,$$

for $\alpha \in (0, 1)$.

• **Solution:** Setting $a \exp\left(-\frac{bw^2}{c + dw}\right) = \alpha$ and solving for w gives

$$\begin{aligned} w &= \frac{1}{2b} \left(d \ln\left(\frac{a}{\alpha}\right) + \sqrt{d^2 \ln^2\left(\frac{a}{\alpha}\right) + 4bc \ln\left(\frac{a}{\alpha}\right)} \right) \\ &\leq \frac{1}{2b} \left(d \ln\left(\frac{a}{\alpha}\right) + \sqrt{d^2 \ln^2\left(\frac{a}{\alpha}\right)} + \sqrt{4bc \ln\left(\frac{a}{\alpha}\right)} \right), \end{aligned}$$

because $\sqrt{a + b} \leq \sqrt{a} + \sqrt{b}$. The result now follows because $\mathbb{P}[|X| \leq w] \leq \mathbb{P}[|X| \leq w']$ for any $w' \geq w$.

4. Assuming that X_i is sub-Gaussian with *known* parameter γ_i for $i = 1, 2, \dots, n$, propose a confidence interval for μ with a coverage probability of at least 95%.

• **Solution:** Using parts 2 and 3, with $a = 2$, $b = n$, $c = 2\tilde{\gamma}^2$ and $d = 0$, we can set

$$I_n = \left[\bar{X} - \sqrt{2 \ln(40)} \frac{\tilde{\gamma}}{\sqrt{n}}, \bar{X} + \sqrt{2 \ln(40)} \frac{\tilde{\gamma}}{\sqrt{n}} \right],$$

to conclude that $\mathbb{P}[\mu \in I_n] \geq 0.95$.

2 Question 2: Statistical Decision Theory (25 points)

Let $\mathbf{X} \sim \mathbb{P}_\theta$ be a random vector with $\mathbb{P}_\theta \in \mathcal{P} = \{\mathbb{P}_\theta : \theta \in \Theta\}$. Assume that $T = T(\mathbf{X})$ is a complete sufficient statistic for $\mathcal{P}$ and that $V = V(\mathbf{X})$ is an ancillary statistic for $\mathcal{P}$.

1. Show that $p_A = \mathbb{E}_\theta[\mathbb{1}(V \in A)]$ and $q_A(T) = \mathbb{E}_\theta[\mathbb{1}(V \in A) | T]$ are not functions of θ , for all Borel measurable sets A .

- **Solution:** $p_A = \mathbb{E}_\theta[\mathbb{1}(V \in A)] = \mathbb{P}_\theta[V \in A]$ is not a function of θ by definition of ancillary. $q_A(T) = \mathbb{E}_\theta[\mathbb{1}(V \in A)|T] = \mathbb{P}_\theta[V \in A|T]$ is not a function of θ by definition of sufficiency.

2. Show that $p_A = \mathbb{E}_\theta[q_A(T)]$, and that $p_A = q_A(T)$ almost surely.

- **Solution:** $p_A = \mathbb{E}_\theta[\mathbb{1}(V \in A)] = \mathbb{E}_\theta[\mathbb{E}_\theta[\mathbb{1}(V \in A)|T]] = \mathbb{E}_\theta[q_A(T)]$. Therefore, $\mathbb{E}_\theta[q_A(T) - p_A] = 0$ and by completeness of T it follows that $\mathbb{P}_\theta[q_A(T) - p_A = 0] = 1$, which gives the result.

3. Show that $\mathbb{P}_\theta(T \in B, V \in A) = \mathbb{P}_\theta(T \in B)\mathbb{P}_\theta(V \in A)$ for all Borel measurable sets A and B .

- **Solution:** Using the previous results,

$$\begin{aligned} \mathbb{P}_\theta(T \in B, V \in A) &= \mathbb{E}_\theta[\mathbb{1}(T \in B)\mathbb{1}(V \in A)] \\ &= \mathbb{E}_\theta[\mathbb{1}(T \in B)\mathbb{E}_\theta(\mathbb{1}(V \in A)|T)] \\ &= \mathbb{E}_\theta[\mathbb{1}(T \in B)q_A(T)] \\ &= \mathbb{E}_\theta[\mathbb{1}(T \in B)p_A] \\ &= \mathbb{E}_\theta[\mathbb{1}(T \in B)]p_A \\ &= \mathbb{P}_\theta(T \in B)\mathbb{P}_\theta(V \in A). \end{aligned}$$

This is known as Basu's theorem.

4. Let $\mathbf{X} = (X_1, \dots, X_n)$ be an i.i.d. random sample from $\text{Normal}(\mu, \sigma^2)$, with $\sigma > 0$ known. Let

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \quad \text{and} \quad \bar{X}(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^n w_i X_i,$$

for some non-random weights $\mathbf{w} = (w_1, \dots, w_n)'$.

- (a) Show that $\bar{X}(\mathbf{w})$ is an ancillary statistic for μ if and only if $\sum_{i=1}^n w_i = 0$.

- **Solution:** Using properties of the normal distribution, it follows that

$$\bar{X}(\mathbf{w}) \sim \text{Normal} \left(\mu \frac{1}{n} \sum_{i=1}^n w_i, \sigma^2 \frac{1}{n} \sum_{i=1}^n w_i^2 \right),$$

which is not a function of μ if and only if $\sum_{i=1}^n w_i = 0$.

(b) Provide sufficient conditions for $\bar{X}$ and $\bar{X}(\mathbf{w})$ to be independent.

- **Solution:** Recall that $\bar{X}$ is a complete sufficient statistic for μ , provided that $\mu \neq \sigma$. Thus, setting $T = \bar{X}$ and $V = \bar{X}(\mathbf{w})$ and by using the result from part 3, we conclude that $\bar{X}$ and $\bar{X}(\mathbf{w})$ are independent if $\sum_{i=1}^n w_i = 0$.

3 Question 3: Finite-Sample Statistical Inference (50 points)

Let $(X_i : 1 \leq i \leq n)$ be an i.i.d. random sample from $X \sim \mathbb{P}_\theta$, with absolutely continuous d.f.

$$F(x; \theta) = \mathbb{P}_\theta(X \leq x) = (a + bx^{-1/\theta}) \mathbb{1}\{x > 2\},$$

where $\theta \in \Theta = (0, 1)$ is an unknown parameter, and a and b do not depend on x .

1. Consider basic distributional properties of $\mathbb{P}_\theta$.

(a) Find the values of a and b .

- **Solution:** Since $F(\cdot, \theta)$ is an absolutely continuous c.d.f., we have $1 = \lim_{x \rightarrow \infty} F(x; \theta) = a$ and $0 = \lim_{x \downarrow 2} F(x; \theta) = a + b2^{-1/\theta}$. This implies $a = 1$ and $b = -2^{1/\theta}$.

(b) Find the Lebesgue density of $\mathbb{P}_\theta$, denoted $f(\cdot; \theta)$. Is $\{\mathbb{P}_\theta : \theta \in \Theta\}$ an exponential family?

- **Solution:** The density is

$$f(x; \theta) = \frac{d}{dx} F_X(x; \theta) = \frac{2^{1/\theta}}{\theta} x^{-(1+\theta)/\theta} \mathbb{1}(x > 2) = h(x)c(\theta) \exp(T(x)w(\theta)),$$

where $h(x) = \mathbb{1}(x > 2)$, $c(\theta) = 2^{1/\theta}/\theta$, $T(x) = \log(x)$ and $w(\theta) = -(1 + \theta)/\theta$, which implies that $\{\mathbb{P}_\theta : \theta \in \Theta\}$ forms an exponential family.

2. Consider point estimation of θ using the following two estimators:

$$\hat{\theta}_{\text{MM}} = \frac{\bar{X} - 2}{\bar{X}} \quad \text{and} \quad \hat{\theta}_{\text{ML}} = \frac{1}{n} \sum_{i=1}^n \log\left(\frac{X_i}{2}\right),$$

where $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$. The estimators $\hat{\theta}_{\text{MM}}$ and $\hat{\theta}_{\text{ML}}$ correspond to the method of moments and maximum likelihood estimators respectively.

(a) Compute $\mathbb{E}_\theta[X_i]$. Characterize the direction of bias (if any) of $\hat{\theta}_{\text{MM}}$: is $\hat{\theta}_{\text{MM}}$ on average an overestimate or underestimate of θ ?

- **Solution:** The expected value is

$$\mathbb{E}_\theta[X_i] = \int_2^\infty x f(x; \theta) dx = \int_2^\infty \frac{2^{1/\theta}}{\theta} x^{-1/\theta} dx = -\frac{2^{1/\theta}}{1 - \theta} x^{-(1-\theta)/\theta} \Big|_{x=2}^\infty = \frac{2}{1 - \theta}.$$

By Jensen's inequality (applied to the convex function $x \mapsto 2/x$),

$$\mathbb{E}_\theta[\hat{\theta}_{\text{MM}}] = 1 - \mathbb{E}_\theta\left[\frac{2}{\bar{X}_n}\right] < 1 - \frac{2}{\mathbb{E}_\theta[\bar{X}_n]} = \theta.$$

This inequality is strict as $x \mapsto 2/x$ is strictly convex and $\bar{X}$ is not a.s. constant. Hence $\hat{\theta}_{\text{MM}}$ underestimates θ on average. That this can also be shown using the Cauchy-Schwarz inequality.

(b) Show that $\log(X_i/2) \sim \text{Exponential}(\theta)$.

Compute the bias of $\hat{\theta}_{\text{ML}}$ for θ . If biased, characterize the direction.

- **Solution:** Let $Y = \log(X/2)$. Then, $\mathbb{P}_\theta(Y \leq y) = \mathbb{P}_\theta(\log(X/2) \leq y) = \mathbb{P}_\theta(X \leq 2 \exp(y))$. This implies that $\mathbb{P}_\theta(Y \leq y) = (1 - \exp(-y/\theta)) \mathbb{1}(y > 0)$ and the Lebesgue density is

$$f_Y(y; \theta) = \frac{1}{\theta} \exp\left(-\frac{y}{\theta}\right) \mathbb{1}(y > 0).$$

which means that $Y \sim \text{Exponential}(\theta)$. From the properties of the exponential distribution, $\mathbb{E}_\theta[\hat{\theta}_{\text{ML}}] = \theta$, and therefore $\hat{\theta}_{\text{ML}}$ is unbiased for θ .

(c) Determine whether $\hat{\theta}_{\text{MM}}$ is based on a complete sufficient statistic.

Determine whether $\hat{\theta}_{\text{ML}}$ is based on a complete sufficient statistic.

- **Solution:** By properties of the exponential family, $T(X_1, \dots, X_n) = \sum_{i=1}^n \log(X_i)$ is a sufficient statistic, and completeness follows from the fact that the parameter space

$$\left\{ -\left(1 + \frac{1}{\theta}\right) : \theta \in (0, 1) \right\} = (-\infty, -2)$$

contains an open set. Clearly, $\hat{\theta}_{\text{MM}}$ is not based on $T(X)$, while $\hat{\theta}_{\text{ML}}$ is. This already verifies that $\hat{\theta}_{\text{ML}}$ is UMVU because it is unbiased.

(d) Compute the Cramér-Rao lower bound for any (regular) unbiased estimator of θ .

Determine whether $\hat{\theta}_{\text{MM}}$ is a UMVU estimator.

Determine whether $\hat{\theta}_{\text{ML}}$ is a UMVU estimator.

In each case, verify whether the Cramér-Rao lower bound is attained.

- **Solution:** The log-likelihood function is

$$\ell_n(\theta) = -\frac{1+\theta}{\theta} \sum_{i=1}^n \log(X_i) - n \log(\theta) + \frac{n}{\theta} \log 2,$$

and therefore

$$\ell_{\theta,n}(\theta) = \frac{d}{d\theta} \ell_n(\theta) = -\frac{n}{\theta} + \frac{1}{\theta^2} \sum_{i=1}^n \log(X_i) - \frac{n}{\theta^2} \log 2 = -\frac{n}{\theta} + \frac{1}{\theta^2} \sum_{i=1}^n \log(X_i/2),$$

and

$$\ell_{\theta\theta,n}(\theta) = \frac{d^2}{d\theta^2} \ell_n(\theta) = \frac{n}{\theta^2} - \frac{2}{\theta^3} \sum_{i=1}^n \log(X_i) + \frac{2n}{\theta^3} \log 2 = \frac{1}{\theta} \frac{d}{dx} \ell_n(\theta) - \frac{1}{\theta^3} \sum_{i=1}^n \log(X_i).$$

The information number is

$$\mathcal{I}(\theta) = \mathbb{V}_\theta[\ell_{\theta,n}(\theta)] = \mathbb{V}_\theta \left[-\frac{n}{\theta} + \frac{1}{\theta^2} \sum_{i=1}^n \log(X_i/2) \right] = \frac{n}{\theta^4} \mathbb{V}_\theta[Y] = \frac{n}{\theta^2},$$

using the fact that $Y \sim \text{Exponential}(\theta)$. Therefore, the CRLB for unbiased estimators of θ is $\text{CRLB}(\theta) = \mathcal{I}(\theta)^{-1} = \theta^2/n$, which is attained by $\hat{\theta}_{\text{ML}}$ because

$$\mathbb{V}_\theta[\hat{\theta}_{\text{ML}}] = \mathbb{V}_\theta \left[\frac{1}{n} \sum_{i=1}^n \log(X_i/2) \right] = \frac{\mathbb{V}_\theta[Y]}{n} = \frac{\theta^2}{n} = \text{CRLB}(\theta).$$

On the other hand, $\hat{\theta}_{\text{MM}}$ is not based on a complete sufficient statistic (nor it is unbiased), which implies that it cannot be an UMVU estimator.

3. Consider the following one-sided hypothesis testing problem:

$$H_0 : \theta \geq \theta_0 \quad \text{vs.} \quad H_1 : \theta < \theta_0, \quad \text{for some fixed } \theta_0 \in (0, 1).$$

(a) Find the uniformly most powerful (UMP) testing procedure of level $\alpha \in (0, 1)$.

- **Solution:** we will use Karlin-Rubin, so we start by checking the MLR condition. Take $\theta_0 < \theta_1$, then

$$\begin{aligned} \frac{f(X, \theta_1)}{f(X, \theta_0)} &= \frac{2^{1/\theta_1}/\theta_1}{2^{1/\theta_0}/\theta_0} \cdot \frac{\exp \{-(1 + 1/\theta_1) \sum_{i=1}^n \log(X_i)\}}{\exp \{-(1 + 1/\theta_0) \sum_{i=1}^n \log(X_i)\}} \\ &= 2^{-(1/\theta_0 - 1/\theta_1)} \frac{\theta_0}{\theta_1} \exp \left\{ (1/\theta_0 - 1/\theta_1) \sum_{i=1}^n \log(X_i) \right\} \end{aligned}$$

which is increasing in $\sum_{i=1}^n \log(X_i)$, and hence in $n^{-1} \sum_{i=1}^n \log(X_i/2)$, so the MLR property is satisfied. Therefore, the UMP test rejects if

$$\frac{1}{n} \sum_{i=1}^n \log(X_i/2) < c \quad \Longleftrightarrow \quad \frac{1}{\theta_0 n} \sum_{i=1}^n \log(X_i/2) = \frac{\hat{\theta}_{\text{ML}}}{\theta_0} < c^*$$

where, by properties of the exponential distribution,

$$\frac{\hat{\theta}_{\text{ML}}}{\theta_0} \sim \text{Gamma}(n, 1/n)$$

which is pivotal. The constant c^* is

$$\mathbb{P}_{\theta_0} \left[\frac{\hat{\theta}_{\text{ML}}}{\theta_0} < c^* \right] = \alpha \quad \Longleftrightarrow \quad c^* = \gamma_{n, 1/n}(\alpha)$$

where $\gamma_{n, 1/n}(\alpha)$ is the α -quantile from a $\text{Gamma}(n, 1/n)$.

(b) Construct an exact $(1 - \alpha)$ confidence interval for θ .

- **Solution:** we have

$$1 - \alpha = \mathbb{P}_\theta \left[\frac{\hat{\theta}_{\text{ML}}}{\theta} \geq \gamma_{n, 1/n} \right] = \mathbb{P}_\theta \left[\theta \leq \frac{\hat{\theta}_{\text{ML}}}{\gamma_{n, 1/n}} \right]$$

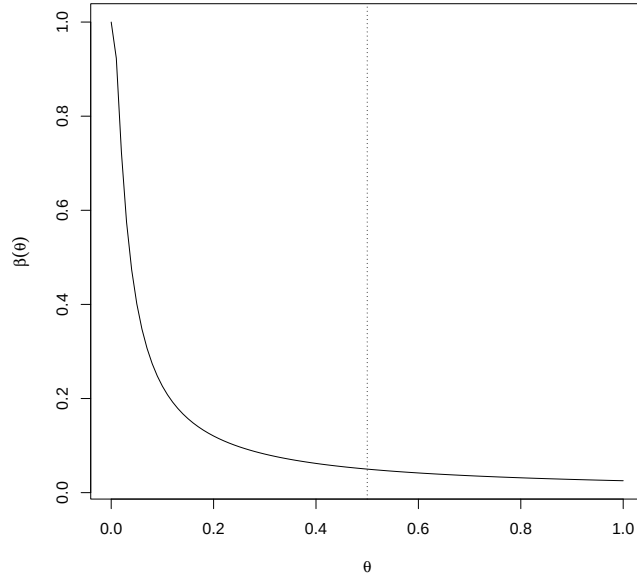


Figure 1: Power function for $\theta_0 = 0.5$, $n = 10$, $\alpha = 0.05$.

so the confidence interval is

$$CI_{n,1-\alpha}(\theta) = \left[0, \frac{\hat{\theta}_{\text{ML}}}{\gamma_{n,1/n}} \right]$$

(c) Find the power function $\beta_n(\theta)$ of the UMP test and sketch its graph.

• **Solution:** The power function is

$$\beta_n(\theta) = \mathbb{P}_\theta \left[\frac{\hat{\theta}_{\text{ML}}}{\theta_0} < \gamma_{n,1/n}(\alpha) \right] = \mathbb{P}_\theta \left[\frac{\hat{\theta}_{\text{ML}}}{\theta} < \frac{\theta_0}{\theta} \gamma_{n,1/n}(\alpha) \right] = \Gamma_{n,1/n} \left(\frac{\theta_0}{\theta} \gamma_{n,1/n}(\alpha) \right),$$

$\theta \in (0, 1)$, where $\Gamma_{n,1/n}(\cdot)$ denotes the c.d.f. of a **Gamma**($n, 1/n$) random variable. The function is plotted in Figure 1. It is easy to see that $\beta_n(0) = 1$ and β_n is decreasing and $\beta_n(\theta_0) = \alpha$.

(d) For each $\theta \in (0, 1)$, find $\lim_{n \rightarrow \infty} \beta_n(\theta)$.

• **Solution:** Let $Y_n \sim \text{Gamma}(n, 1/n)$. By definition of $\gamma_{n,1/n}(\alpha)$ and the hint,

$$\begin{aligned} \alpha &= \mathbb{P}(Y_n \leq \gamma_{n,1/n}(\alpha)) \\ &= \mathbb{P}(\sqrt{n}(Y_n - 1) \leq \sqrt{n}(\gamma_{n,1/n}(\alpha) - 1)) \\ &\rightarrow \Phi \left(\lim_{n \rightarrow \infty} \sqrt{n}(\gamma_{n,1/n}(\alpha) - 1) \right), \end{aligned}$$

which implies that $\sqrt{n}(\gamma_{n,1/n}(\alpha) - 1) \rightarrow z_\alpha$, the α -quantile of a standard normal.

Therefore

$$\begin{aligned}
\beta_n(\theta) &= \Gamma_{n,1/n} \left(\frac{\theta_0}{\theta} \gamma_{n,1/n}(\alpha) \right) \\
&= \mathbb{P} \left(Y_n \leq \frac{\theta_0}{\theta} \gamma_{n,1/n}(\alpha) \right) \\
&= \mathbb{P} \left(\sqrt{n}(Y_n - 1) \leq \sqrt{n} \left(\frac{\theta_0}{\theta} \gamma_{n,1/n}(\alpha) - 1 \right) \right) \\
&= \mathbb{P} \left(\sqrt{n}(Y_n - 1) \leq \sqrt{n} \frac{\theta_0}{\theta} (\gamma_{n,1/n}(\alpha) - 1) + \sqrt{n} \left(\frac{\theta_0}{\theta} - 1 \right) \right) \\
&\rightarrow \Phi \left(\frac{\theta_0}{\theta} z_\alpha + \lim_{n \rightarrow \infty} \sqrt{n} \left(\frac{\theta_0}{\theta} - 1 \right) \right) \\
&= \begin{cases} \Phi(z_\alpha) = \alpha & \theta = \theta_0 \\ \Phi(\infty) = 1 & \theta < \theta_0 \\ \Phi(-\infty) = 0 & \theta > \theta_0 \end{cases}
\end{aligned}$$

Hint: You may use the following two facts without proof.

- (i) If $W_1, \dots, W_n$ are i.i.d. $\text{Exponential}(\lambda)$, then $\sum_{i=1}^n W_i \sim \text{Gamma}(n, \lambda)$.
- (ii) If $Y_n \sim \text{Gamma}(n, \lambda_n)$, then $\lim_{n \rightarrow \infty} \mathbb{P} \left(\frac{Y_n - n\lambda_n}{\sqrt{n\lambda_n}} \leq x \right) = \Phi(x)$ for all $x \in \mathbb{R}$, where Φ is the c.d.f. of the standard normal distribution.

Table 1.1. Discrete Distributions on $\mathcal{R}$

Uniform	p.d.f.	$1/m, x = a_1, \dots, a_m$
	m.g.f.	$\sum_{j=1}^m e^{a_j t}/m, t \in \mathcal{R}$
$DU(a_1, \dots, a_m)$	Expectation	$\sum_{j=1}^m a_j/m$
	Variance	$\sum_{j=1}^m (a_j - \bar{a})^2/m, \bar{a} = \sum_{j=1}^m a_j/m$
	Parameter	$a_i \in \mathcal{R}, m = 1, 2, \dots$
Binomial	p.d.f.	$\binom{n}{x} p^x (1-p)^{n-x}, x = 0, 1, \dots, n$
	m.g.f.	$(pe^t + 1 - p)^n, t \in \mathcal{R}$
$Bi(p, n)$	Expectation	np
	Variance	$np(1-p)$
	Parameter	$p \in [0, 1], n = 1, 2, \dots$
Poisson	p.d.f.	$\theta^x e^{-\theta}/x!, x = 0, 1, 2, \dots$
	m.g.f.	$e^{\theta(e^t - 1)}, t \in \mathcal{R}$
$P(\theta)$	Expectation	θ
	Variance	θ
	Parameter	$\theta > 0$
Geometric	p.d.f.	$(1-p)^{x-1}p, x = 1, 2, \dots$
	m.g.f.	$pe^t/[1 - (1-p)e^t], t < -\log(1-p)$
$G(p)$	Expectation	$1/p$
	Variance	$(1-p)/p^2$
	Parameter	$p \in [0, 1]$
Hyper-geometric	p.d.f.	$\binom{n}{x} \binom{m}{r-x} / \binom{N}{r}$
		$x = 0, 1, \dots, \min\{r, n\}, r - x \leq m$
$HG(r, n, m)$	m.g.f.	No explicit form
	Expectation	rn/N
	Variance	$rn m(N-r)/[N^2(N-1)]$
	Parameter	$r, n, m = 1, 2, \dots, N = n + m$
Negative binomial	p.d.f.	$\binom{x-1}{r-1} p^r (1-p)^{x-r}, x = r, r+1, \dots$
	m.g.f.	$p^r e^{rt}/[1 - (1-p)e^t]^r, t < -\log(1-p)$
$NB(p, r)$	Expectation	r/p
	Variance	$r(1-p)/p^2$
	Parameter	$p \in [0, 1], r = 1, 2, \dots$
Log-distribution	p.d.f.	$-(\log p)^{-1} x^{-1} (1-p)^x, x = 1, 2, \dots$
	m.g.f.	$\log[1 - (1-p)e^t]/\log p, t \in \mathcal{R}$
$L(p)$	Expectation	$-(1-p)/(p \log p)$
	Variance	$-(1-p)[1 + (1-p)/\log p]/(p^2 \log p)$
	Parameter	$p \in (0, 1)$

All p.d.f.'s are w.r.t. counting measure.

Table 1.2. Distributions on $\mathcal{R}$ with Lebesgue p.d.f.'s

Uniform	p.d.f.	$(b-a)^{-1}I_{(a,b)}(x)$
	m.g.f.	$(e^{bt} - e^{at})/[(b-a)t], \quad t \in \mathcal{R}$
$U(a, b)$	Expectation	$(a+b)/2$
	Variance	$(b-a)^2/12$
	Parameter	$a, b \in \mathcal{R}, \quad a < b$
Normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}e^{-(x-\mu)^2/2\sigma^2}$
	m.g.f.	$e^{\mu t + \sigma^2 t^2/2}, \quad t \in \mathcal{R}$
$N(\mu, \sigma^2)$	Expectation	μ
	Variance	σ^2
	Parameter	$\mu \in \mathcal{R}, \quad \sigma > 0$
Exponential	p.d.f.	$\theta^{-1}e^{-(x-a)/\theta}I_{(a,\infty)}(x)$
	m.g.f.	$e^{at}(1-\theta t)^{-1}, \quad t < \theta^{-1}$
$E(a, \theta)$	Expectation	$\theta + a$
	Variance	θ^2
	Parameter	$\theta > 0, \quad a \in \mathcal{R}$
Chi-square	p.d.f.	$\frac{1}{\Gamma(k/2)2^{k/2}}x^{k/2-1}e^{-x/2}I_{(0,\infty)}(x)$
	m.g.f.	$(1-2t)^{-k/2}, \quad t < 1/2$
χ_k^2	Expectation	k
	Variance	$2k$
	Parameter	$k = 1, 2, \dots$
Gamma	p.d.f.	$\frac{1}{\Gamma(\alpha)\gamma^\alpha}x^{\alpha-1}e^{-x/\gamma}I_{(0,\infty)}(x)$
	m.g.f.	$(1-\gamma t)^{-\alpha}, \quad t < \gamma^{-1}$
$\Gamma(\alpha, \gamma)$	Expectation	$\alpha\gamma$
	Variance	$\alpha\gamma^2$
	Parameter	$\gamma > 0, \quad \alpha > 0$
Beta	p.d.f.	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}I_{(0,1)}(x)$
	m.g.f.	No explicit form
$B(\alpha, \beta)$	Expectation	$\alpha/(\alpha+\beta)$
	Variance	$\alpha\beta/[(\alpha+\beta+1)(\alpha+\beta)^2]$
	Parameter	$\alpha > 0, \quad \beta > 0$
Cauchy	p.d.f.	$\frac{1}{\pi\sigma}\left[1+\left(\frac{x-\mu}{\sigma}\right)^2\right]^{-1}$
	ch.f.	$e^{\sqrt{-1}\mu t - \sigma t }$
$C(\mu, \sigma)$	Expectation	Does not exist
	Variance	Does not exist
	Parameter	$\mu \in \mathcal{R}, \quad \sigma > 0$

Table 1.2. (continued)

t-distribution	p.d.f.	$\frac{\Gamma[(n+1)/2]}{\sqrt{n\pi}\Gamma(n/2)} \left(1 + \frac{x^2}{n}\right)^{-(n+1)/2}$
	ch.f.	No explicit form
t_n	Expectation	0, ($n > 1$)
	Variance	$n/(n-2)$, ($n > 2$)
	Parameter	$n = 1, 2, \dots$
F-distribution	p.d.f.	$\frac{n^{n/2}m^{m/2}\Gamma[(n+m)/2]x^{n/2-1}}{\Gamma(n/2)\Gamma(m/2)(m+nx)^{(n+m)/2}}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$F_{n,m}$	Expectation	$m/(m-2)$, ($m > 2$)
	Variance	$2m^2(n+m-2)/[n(m-2)^2(m-4)]$, ($m > 4$)
	Parameter	$n = 1, 2, \dots$, $m = 1, 2, \dots$
Log-normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}x^{-1}e^{-(\log x - \mu)^2/2\sigma^2}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$LN(\mu, \sigma^2)$	Expectation	$e^{\mu+\sigma^2/2}$
	Variance	$e^{2\mu+\sigma^2}(e^{\sigma^2} - 1)$
	Parameter	$\mu \in \mathcal{R}$, $\sigma > 0$
Weibull	p.d.f.	$\frac{\alpha}{\theta}x^{\alpha-1}e^{-x^\alpha/\theta}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$W(\alpha, \theta)$	Expectation	$\theta^{1/\alpha}\Gamma(\alpha^{-1} + 1)$
	Variance	$\theta^{2/\alpha} \left\{ \Gamma(2\alpha^{-1} + 1) - [\Gamma(\alpha^{-1} + 1)]^2 \right\}$
	Parameter	$\theta > 0$, $\alpha > 0$
Double Exponential	p.d.f.	$\frac{1}{2\theta}e^{- x-\mu /\theta}$
	m.g.f.	$e^{\mu t}/(1 - \theta^2 t^2)$, $ t < \theta^{-1}$
$DE(\mu, \theta)$	Expectation	μ
	Variance	$2\theta^2$
	Parameter	$\mu \in \mathcal{R}$, $\theta > 0$
Pareto	p.d.f.	$\theta a^\theta x^{-(\theta+1)}I_{(a,\infty)}(x)$
	ch.f.	No explicit form
$Pa(a, \theta)$	Expectation	$\theta a/(\theta - 1)$, ($\theta > 1$)
	Variance	$\theta a^2/[(\theta - 1)^2(\theta - 2)]$, ($\theta > 2$)
	Parameter	$\theta > 0$, $a > 0$
Logistic	p.d.f.	$\sigma^{-1}e^{-(x-\mu)/\sigma}/[1 + e^{-(x-\mu)/\sigma}]^2$
	m.g.f.	$e^{\mu t}\Gamma(1 + \sigma t)\Gamma(1 - \sigma t)$, $ t < \sigma^{-1}$
$LG(\mu, \sigma)$	Expectation	μ
	Variance	$\sigma^2\pi^2/3$
	Parameter	$\mu \in \mathcal{R}$, $\sigma > 0$